CONFIDENTIAL

School of Computing,
Faculty of Engineering

FINAL EXAMINATION
SEMESTER 2, 2019/2020

PART B1

SUBJECT CODE	:	**SCSR2043**
SUBJECT	:	**OPERATING SYSTEM**
YEARS / COURSE	:	**SCSI/J/R/V/D/B**
TIME	:	**70 MINUTES**
DATE	:	**15 JULY 2020**
PLACE	:	**ONLINE**

INSTRUCTIONS :

1. Answer **ALL** questions by writing on empty A4 paper;
2. For each page of your answer, write your **NAME**, **MATRIC NO.**, and **SECTION** of the subject (a template of answer sheet is provided separately);
3. Do snapshot or scan each page, convert into pdf, and compile as a single pdf file;
4. Upload the pdf file using the link provided in the e-learning.

This questions paper consists of **3** (**THREE**) printed pages, excluding this page.

PART B1

STRUCTURED QUESTIONS (35 MARKS)

QUESTION 1 [20 *Marks*]

A paging system with an *X*-bytes of logical memory with some portions of logical addresses (decimal) with its contents as shown in Table 1 need to be mapped to an *Y*-bytes of physical memory based on the complete 2^3 entries for a page table shown in Table 2. Assume that each entry size is 2^2-bytes and the physical memory size is 2^6 frames. Answer the following questions.

Table 1

Logical Address	Content
0	B
1	-
2	-
3	A
4	D
5	G
. . .	
19	E
20	-
21	H
22	C
23	F
. . .	

Table 2

Page#	Frame#
0	20
1	45
2	5
3	12
4	10
5	9
6	30
7	4

(a) Calculate the size of *X* and *Y*. Show your working. [4 *Marks*]

(b) What will be the contents of frame 9? Justify your answer. [3 *Marks*]

(c) How many bits will be used for the page number and frame number in logical memory address and physical memory address, respectively? Show your working. [4 *Marks*]

(d) Calculate the logical address (decimal) given a physical address as 10110110_2. Show your working. [4 *Marks*]

(e) Calculate the physical address of data ***E*** in binary of 8 bits. Assume that the initial offset is 0. Show your working. [3 *Marks*]

(f) Write TWO consequences if the paging system resize the frame to 2^3-bytes? Justify each of your answer. [2 *Marks*]

QUESTION 2 [15 Marks]

(a) Consider a memory contains page of 100 bytes each and the addresses space as shown in Figure 1.

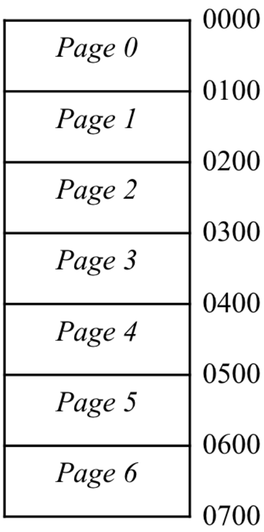

Figure 1

→	0100	0321	0555	0111
	0400	0350	0100	0280
	0050	0050	0010	0080
	0603	0509	0455	0300
	0030	0100	0200	0330
	0010	0520	0100	0001

Figure 2

i) If CPU accesses the addresses shown in Figure 2, write the complete reference string. (Note: The arrow indicates the starting address from the first row). [2 *Marks*]

ii) If the system using any policy of the page replacement with 7 frames for the reference string above, what will be the consequence to the page faults? Briefly justify your answer. [3 *Marks*]

(b) Table 3 illustrates the frames used with the current page reference using LRU page replacement algorithms at time t_{10}. Answer the following questions.

Table 3

Frame	0	1	2	3
Page Reference	2	1	3	4

i) Consider the following new page reference string:

2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 5

By constructing a table with the page reference in the frames, identify how many page faults would occur using the same page replacement algorithms? Show your working. [7 *Marks*]

ii) Based on the previous answer, how to reduce the page fault number? Briefly, justify your answer. [3 *Marks*]

------- Continue in Part B2 -------